TILINGS OF W_M AND T_M BELOW THE THRESHOLD SCALE: A SIX-PIECE LAYOUT

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Abstract. Let $e < f$ be coprime integers. The triangle with sides ef, $f^2 - e^2$, f^2 (the *tile*) has a family of larger triangles W_M, $M = 1, 2, \ldots$, whose area is $M^2(2f^2 - e^2)$ tile areas. Paliy showed that W_M is tiled by copies of the tile when $M \geq f$. We show that W_M is already tiled when $Me \geq f$ and $(M-1)(f-e) \geq f$ (for $2 \leq e \leq f-2$). The tiling is one explicit layout of six convex pieces; to prove that the pieces fill W_M without overlap it suffices to check that the 22 explicit cross products of the pieces (and 3 of the target), which factor into products of nine simple expressions, are positive. The smallest example is a triangle with sides $315, 276, 375$ tiled by 414 copies of the triangle $(10, 21, 25)$. The statement is a theorem of a Lean 4 library, checked by the Lean kernel inside this project (no outside check); the hand proof here is a second, unformalised proof of the same layout. That no smaller scale works is proved only for $Me < f$, and only by importing a theorem of an earlier release. The paper says, for each statement, whether it is Lean, hand, computed or imported.

1. The result at a glance

Tilings. A triangle is *tiled by N copies of a tile* if it is the union of N triangles with pairwise disjoint interiors, each congruent to the tile (mirror images allowed, no condition that edges match). Erdős asked which integers N occur (Problem 634 of `erdosproblems.com`); this paper only adds tilings, it says nothing about primes.

An example. Take the tile with sides 10, 21, 25. A *scale* is a whole number s; a triangle with sides $10s, 21s, 25s$ is a tile enlarged s times and is tiled by s^2 tiles. The target of scale 3 is a different triangle, with sides 315, 276, 375. Its area is 414 tiles, and we show that it *is* tiled by 414 copies of the tile. The earlier theorems of Paliy start at scale 5 for this tile.

How it is cut (Figure 1). Call A the corner where the sides 276 and 375 meet and O the corner where 375 and 315 meet. Mark the point on the side 276 that is 150 from A; it is 315 from O, as far from O as the side 315 is long. Joining it to O cuts the target in two. The part next to the side 375 is one big triangle of scale 15 minus a small corner: piece 0 (221 tiles). The other part is a thin triangle; its right part is piece 1, a triangle of scale 11 (121 tiles), and a strip along the left side, one tile-row wide, is filled by four small pieces $2, 3, 4, 5$ ($25 + 9 + 30 + 8$ tiles). Each piece is a triangle, a triangle with a corner cut off, or a parallelogram, and is cut into tiles by lines parallel to its sides. So $221 + 121 + 25 + 9 + 30 + 8 = 414$.

The general statement (Theorem 2.1, Section 2). The tile $(10, 21, 25)$ is $(ef, f^2 - e^2, f^2)$ for $(e, f) = (2, 5)$, and the scale is $M = 3$. For any coprime $2 \leq e \leq f - 2$ the same six pieces tile the corresponding target as soon as $Me \geq f$ and $(M-1)(f-e) \geq f$; for $e < f/2$ the first inequality alone suffices. Further instances (M is the smallest scale covered; the last column is Paliy's smallest scale f):

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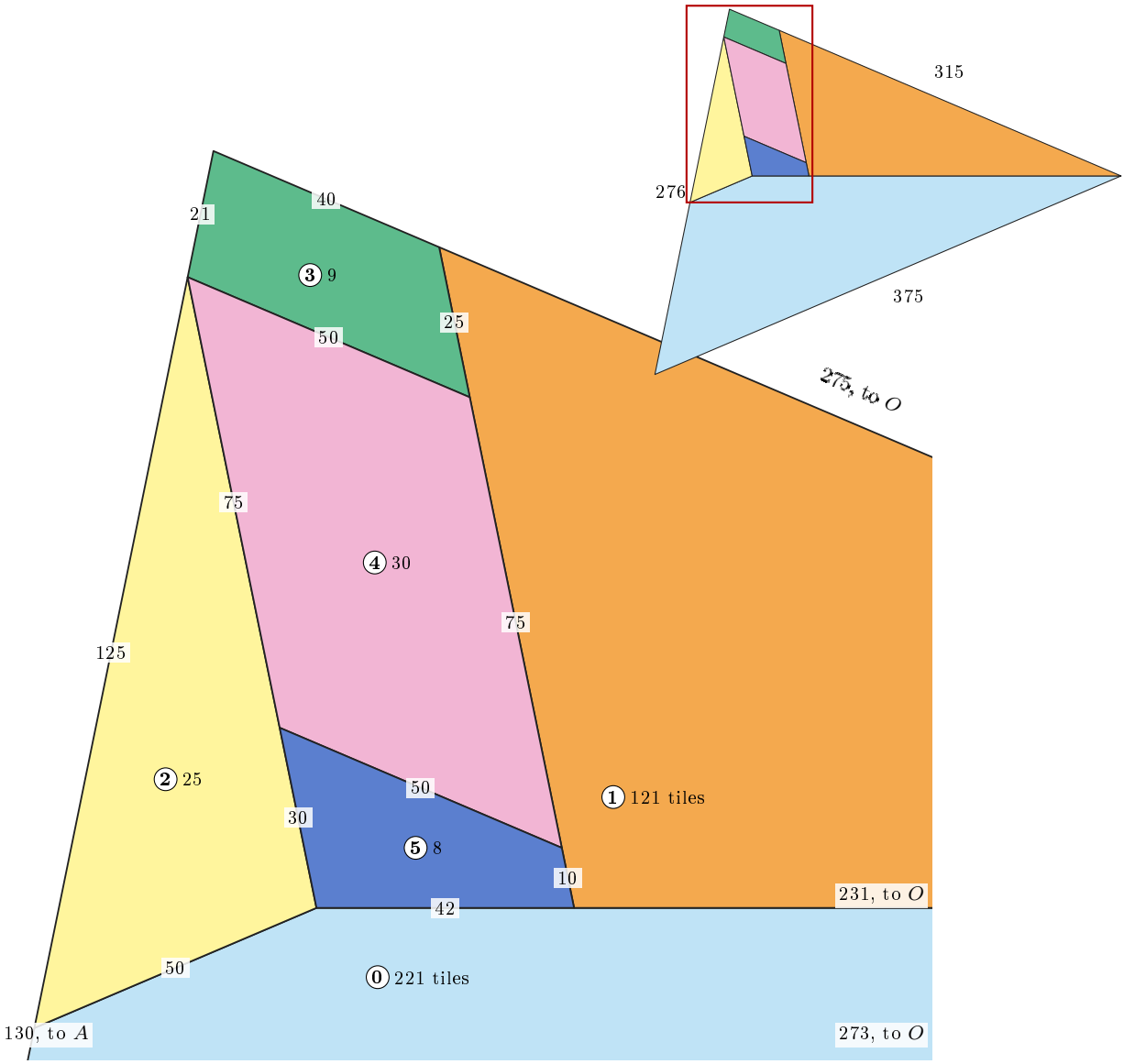


FIGURE 1. The six pieces of the target with sides $315, 276, 375$ (tile $(10, 21, 25)$, 414 tiles). Large panel: the part near the left side, true shape, every edge length written as a number; an arrow with a number means the edge leaves the panel and has that length. Small inset: the whole target; the red box is the large panel. Pieces are numbered 0–5 (number and tile count inside); each piece is cut into tiles by lines parallel to its sides. The four small pieces form a strip; on the right side the numbers $25 + 75 + 10 = 110$ add up to the side of piece 1.

tile (e, f)	sides of the tile	scale M	tiles	scale before
$(2, 5)$	$10, 21, 25$	3	414	5
$(3, 5)$	$15, 16, 25$	4	656	5
$(3, 7)$	$21, 40, 49$	3	801	7
$(2, 11)$	$22, 117, 121$	6	8568	11
$(9, 11)$	$99, 40, 121$	7	7889	11

TABLE 1. What is proved, and how.

statement	status
the target is tiled for the scales above	Lean (kernel-checked, inside this project) and hand proof in Section 4
a neighbouring isosceles target is tiled	hand proof of a few lines (Section 2.2), which also uses the ladder subdivision (Lean, earlier release); the gluing is Lean in an earlier release
no smaller scale works, for $e < f/2$	only the case $Me < f$, from an earlier release (imported, Lean there)
explicit tilings, e.g. the 414 tiles	exact computer check, not Lean
26 138 integers become tile counts	computer count, relative to this project's census

What this does not do. It does not decide which scales fail when $e > f/2$, it does not treat $e = 1$ and $e = f - 1$, and it says nothing about primes. No reader outside this project has checked any of it. Section 2 states exactly what is Lean and what is imported; Section 3 (one page) fixes notation; Section 4 is the proof; Section 5 works the example by hand.

2. STATEMENT, AND WHAT IS PROVED WHERE

Throughout, $e < f$ are coprime positive integers, $\lfloor x \rfloor$ and $\lceil x \rceil$ are the floor and the ceiling of x, and

$$a = ef, \quad b = f^2 - e^2, \quad c = f^2, \quad \kappa = b + c = 2f^2 - e^2, \quad d = f - e.$$

The *tile* $\mathsf{tile}(e, f)$ has sides a, b, c; its angles α, β, γ are opposite a, b, c, and $3\alpha{+}2\beta = \pi$, $\gamma = 2\alpha{+}\beta$. For a whole number $M \geq 1$ the two targets are

- $W_M(e, f)$: the triangle with sides Mfb, $Me\kappa$, Mf^3 (angles β, 2α, $\alpha{+}\beta$ opposite them); its area is $M^2(2f^2 - e^2)$ tile areas;
- $T_M(e, f)$: the isosceles triangle with legs Mf^3 and base $Me(3f^2 - e^2)$ (apex 3α, base angles β); area $M^2(3f^2 - e^2)$ tile areas.

Both were introduced by Beeson–Roşu and Paliy (Section 7). For $(e, f, M) = (2, 5, 3)$: W_3 has sides $315, 276, 375$, as in Section 1.

Theorem 2.1 (Theorem W). *Let* $\gcd(e, f) = 1$, $2 \leq e \leq f - 2$, *and let* $M \geq 1$ *satisfy*

$$\textit{(H1)} \quad Me \geq f, \qquad\qquad \textit{(H2)} \quad (M - 1)(f - e) \geq f.$$

Then $W_M(e, f)$ *is tiled by* $M^2(2f^2 - e^2)$ *copies of* $\mathsf{tile}(e, f)$, *and* $T_M(e, f)$ *is tiled by* $M^2(3f^2 - e^2)$ *copies.*

Remark 2.2 (an equivalent form). (H1) and (H2) hold together if and only if $(f - e)\lfloor eM/f \rfloor \geq e + 1$ (Lemma 4.4); equivalently $M \geq M^* := \max(\lceil f/e \rceil, \lceil f/(f - e) \rceil + 1)$.

2.1. Where each part is proved.

(1) **The W half: Lean** [PROVED IN LEAN]. Module `Draft/ThmW.lean` of the project's Lean 4 library (19 helper modules, aggregate `Erdos634.AllThmW`), theorem `Draft.ThmW.thmW`.

For natural numbers e, f, M with `Nat.Coprime e f`, $2 \le e$, $e+2 \le f$, $f \le Me$, $f \le (M-1)(f-e)$ it concludes

$$\texttt{WTilable e f M} \wedge \texttt{TileCount}\big((2f^2-e^2)M^2\big) \wedge \texttt{TileCount}\big(M^2(3f^2-e^2)\big).$$

`WTilable` says: a triangle with sides $Mf(f^2-e^2)$, $Me(2f^2-e^2)$, Mf^3 is dissected into triangles each congruent to the triangle with sides ef, f^2-e^2, f^2. Axioms: `propext`, `Classical.choice`, `Quot.sound`; no `sorry`. The hypotheses are witnessed by instances $(2,5,3)$, $(3,5,4)$, $(3,7,3)$, so the theorem is not vacuous. The Lean statement was compared with the one above inside this project (verdict: match). That Lean proof verifies the same layout by a different route (a table of about 100 cross products in complex coordinates, produced by a program, not meant for reading; it is described in the verification record [7]); it does not follow Section 4. We give the hand proof so that a person can check the layout.

(2) **The W half: by hand** [PROVED BY HAND]. Section 4. This is a second proof of the same layout, written for a reader, with the placement obtained from a chain lemma. It is not formalised: the Lean modules `Draft/ChainKap`, `ChainTri`, `ChainCover`, `ChainW`, `ChainParams` (project repository, no `sorry`, *unaudited*) prove the chain lemma for triangles, the placement of the six pieces split into ten triangles, and Lemma 4.4; they are *not* yet linked to the packing statement, so no Lean theorem `WTilable` is derived from them. The tiling of each piece by tiles (Section 4.3) is not in those modules.

(3) **The T half:** the *count* $M^2(3f^2-e^2)$ is in the Lean theorem above; the *tiling* of T_M is the gluing of Section 2.2, proved there by hand (Lean version in an earlier release, `CollarG.betaTilable_of_wTilable`).

(4) **Imported** (Section 2.3): only for the sharpness corollary and the open part; the proofs of the Theorem use none of them.

2.2. The T half.

Proposition 2.3 (gluing; [PROVED BY HAND], Lean in an earlier release)**.** *If $W_M(e,f)$ is tiled by copies of* $\mathsf{tile}(e,f)$*, so is $T_M(e,f)$, with $M^2(2f^2-e^2)+(Mf)^2 = M^2(3f^2-e^2)$ tiles.*

Proof. Name the vertices of W_M so that $OB = Mfb$, $OA = Mf^3$, $AB = Me\kappa$; the angles are β at A, 2α at O, $\alpha+\beta$ at B. Attach a tile of scale Mf (sides $Mf\cdot a$, $Mf\cdot b$, $Mf\cdot c$) on the side OB outside W_M, its side $Mfb = OB$ on OB, with its angle γ at B and its angle α at O. At B the angles add to $(\alpha+\beta)+\gamma = \pi$, so the third vertex Z lies on the line AB beyond B; at O the angles add to $2\alpha+\alpha = 3\alpha$. The tile has $BZ = Mf\cdot a = Mef^2$ (opposite α) and $OZ = Mf\cdot c = Mf^3 = OA$ (opposite γ and adjacent to α at O). So the union is the triangle OAZ with $OA = OZ = Mf^3$, $AZ = Me\kappa + Mef^2 = Me(3f^2-e^2)$: this is T_M. The extra tile of scale Mf is tiled by $(Mf)^2$ tiles (ladder, Proposition 4.3). □

The converse is not known. The Lean theorem also contains the count for T_M by a statement of the earlier release (`DebtB.counts_of_wTilable`), which needs besides the tiling only $M(f-e) \ge f$; (H2) implies it.

2.3. Imported facts (earlier release).

These are Lean theorems of the earlier releases [4, 5], used as stated, not reproved; *tiled* means by copies of $\mathsf{tile}(e,f)$ with the same (e,f).

(I1) (`cor:ebelow`) For coprime $1 \le e < f$: if $Me < f$ then $W_M(e,f)$ is not tiled; if $2Me < f$ then $T_M(e,f)$ is not tiled.

(I2) (`cor:thickw`(a)) If $M(f-e) \le e$ then $W_M(e,f)$ is not tiled.

(I3) (`cor:below`, `cor:thickw`(b)) For $e = 1$ and for $e = f - 1$: W_M is tiled if and only if $M \geq f$.

Context, not used: Paliy proved that W_M and T_M are tiled for every $M \geq f$ [3].

Corollary 2.4 (sharpness for $e < f/2$; "if" from Theorem 2.1, "only if" imported (I1))**.** *Let* $\gcd(e, f) = 1$, $2 \leq e < f/2$. *Then* $W_M(e, f)$ *is tiled if and only if* $Me \geq f$.

Proof. "Only if" is (I1). "If": $e < f/2$ gives $f/(f-e) < 2$, hence $\lceil f/(f-e) \rceil = 2$ (as $f/(f-e) > 1$) and $\lceil f/(f-e) \rceil + 1 = 3$, while $f/e > 2$ gives $\lceil f/e \rceil \geq 3$; so $M^* = \lceil f/e \rceil$. Also (H1) implies $M \geq 3$ and then $(M-1)(f-e) \geq 2(f-e) > f$, i.e. (H2). Apply Theorem 2.1. □

So for $e < f/2$ the threshold is exactly $\lceil f/e \rceil$, below f as soon as $e \geq 2$; the "only if" half is only as strong as (I1).

3. The frame and the notation

Table 2. Notation, in order of appearance in Sections 2–4.

$e < f$, $d = f - e$	coprime integers defining the tile; $2 \leq e \leq f-2$ in Theorem 2.1, so $d \geq 2$
a, b, c, κ	ef, $f^2 - e^2$, f^2, $2f^2 - e^2$: tile sides and $b + c$
scale s	a triangle with sides sa, sb, sc (an enlarged tile); tiled by s^2 tiles
W_M, T_M, M	the targets of Section 2; M is the target's scale
n, ρ, δ	$n = \lfloor eM/f \rfloor$, $\rho = eM - fn$, $\delta = 1$ if $\rho = 0$ and $\delta = 0$ otherwise
t, r, q	integers of the layout, $(t, r, q) = (n - \delta,\ n + 1,\ n)$ (*canonical*); (K) of Section 4.1
P, O, B, A	$W_M = OAB$, $OB = Mfb$, $OA = Mf^3$, $AB = Me\kappa$; $P \in AB$ with $OP = OB$
(ξ, y)	coordinates in the frame with origin P and axes along PO, PB
v, s_1, s_2, x, u_1, u_2	scales of the pieces, (2)
$\Pi_0, \ldots, \Pi_5$	the six pieces (Table 4)

3.1. **The frame.** Let P be the point of AB with $OP = OB$. Then $PA = Mef^2$ and $PB = AB - PA = Meb$. In the isosceles triangle OPB the angle at P equals the angle at B, with cosine $\frac{PB/2}{OB} = \frac{e}{2f}$. Use P as origin, unit vectors along PO and PB as axes. A point is (ξ, y), and by the law of cosines a vector (ξ, y) has

$$\text{length}^2 = \xi^2 + y^2 + \tfrac{e}{f}\,\xi y, \qquad \text{inner product } \langle (\xi, y), (\xi', y') \rangle = \xi\xi' + yy' + \tfrac{e}{2f}(\xi y' + y\xi'). \tag{1}$$

Then $O = (fbM, 0)$, $B = (0, ebM)$, $A = (0, -ef^2M)$ and (1) gives $|OB| = fbM$, $|OA|^2 = M^2f^2(b^2 + e^2f^2 + e^2b) = (Mf^3)^2$ (as $b^2 + e^2b + e^2f^2 = f^4$), $|AB| = Me\kappa$: this is W_M. Five directions occur; their primitive integer vectors and lengths, by (1), are:

direction	vector	length	used by
PO	$(1, 0)$	1	OP, PP_2, P_2C', $C'X_4$
PB	$(0, 1)$	1	the side AB
$(f, -e)$	$(f, -e)$	f	OB, QV_2, U_1U_2
$(e, -f)$	$(e, -f)$	f	the lines ℓ_k below, QU_1, V_2U_2, P_2U_1
(b, ef)	(b, ef)	f^2	OA, P_2P_3

(Checks: $f^2+e^2-e^2=f^2$ for the third, $e^2+f^2-e^2=f^2$ for the fourth, $b^2+e^2f^2+e^2b=f^4$ for the fifth.) A segment in one of these directions with integer coordinates has length (its integer multiple) $\times$ (the length in the table). *Lattice lines:* with $\varphi=f\xi+ey$, the line $\ell_k=\{\varphi=efbk\}$ has direction $(e,-f)$, meets PO at $\xi=ebk$ and AB at $y=fbk$.

3.2. **The points of the layout.** The layout has three integers t,r,q (Section 4.1; for $W_3(2,5)$ they are $1,2,1$) and six integer scales, which are the sizes of the triangles used to build the pieces (v: piece 1; s_1,s_2: the big triangle and the removed corner of piece 3; x: the second side of the parallelogram, piece 4; u_1,u_2: the big triangle and the removed corner of piece 5; Table 4):

$$v=fM-er,\quad s_1=f(r-t),\quad s_2=fr-eM,\quad x=ft-eq,\quad u_1=fq-et,\quad u_2=fq-er. \tag{2}$$

TABLE 3. The points, in the frame (ξ,y); all coordinates are integers. R and X_4 are not vertices of any piece: they are the apexes of the two triangles from which a corner is cut to make Π_3 and Π_5.

$P=(0,0)$	$O=(fbM,\,0)$	$B=(0,\,ebM)$	$A=(0,\,-ef^2M)$
$P_2=(bet,\,0)$	$P_3=(0,\,-e^2ft)$	$Q=(0,\,fbt)$	$R=(0,\,fbr)$
$C'=(ber,\,0)$	$X_4=(fbq,\,0)$	$B'=\big(ef(fr-eM),\,ef(fM-er)\big)$	
$V_2=\big(ef^2(r-t),\,f(f^2t-e^2r)\big)$		$U_1=\big(ef(ft-eq),\,ef(fq-et)\big)$	
$U_2=\big(ef(fr-eq),\,ef(fq-er)\big)$			

TABLE 4. The six pieces: counter-clockwise vertex lists, shape, and tile count (Section 4.3).

	vertices	shape	tiles
Π_0	A,O,P_2,P_3	triangle OAP of scale fM minus corner PP_2P_3 of scale et	$f^2M^2-e^2t^2$
Π_1	O,B',C'	triangle of scale v	v^2
Π_2	P_3,P_2,Q	triangle of scale ft	f^2t^2
Π_3	B,Q,V_2,B'	triangle RQV_2 of scale s_1 minus corner RBB' of scale s_2	$s_1^2-s_2^2$
Π_4	Q,U_1,U_2,V_2	parallelogram, sides as_1 and cx, angle β	$2s_1x$
Π_5	U_1,P_2,C',U_2	triangle $X_4U_1P_2$ of scale u_1 minus corner $X_4C'U_2$ of scale u_2	$u_1^2-u_2^2$

4. Proof of the W half of Theorem 2.1

Standing assumptions: $\gcd(e,f)=1$, $2\le e\le f-2$, $M\ge1$; t,r,q integers. The proof has three steps: the six pieces fill W_M without overlap (Step 1: chain lemma and nine factors); each piece is a union of tiles (Step 2: 24 edge-length identities); the nine factors are positive for the canonical (t,r,q) (Step 3: Lemma 4.4 and non-negative certificates).

4.1. **The inequalities (K).**

$$\text{(K)}\qquad ft<eM,\qquad eM<fr,\qquad er<fq,\qquad eq<ft. \tag{3}$$

Write $C_3=eM-ft$, $C_4=fr-eM$, $C_6=fq-er$, $C_7=ft-eq$, $C_5=fM-er$ (so (K) says $C_3,C_4,C_6,C_7>0$). Under (K) the following are positive (nine numbers, each from (K) in one line):

$$r>\tfrac{eM}{f}>0,\ q>\tfrac{er}{f}>0,\ t>\tfrac{eq}{f}>0;\qquad f(r-t)=C_3+C_4>0;$$
$$eC_5=fC_3+fC_7+eC_6>0;$$
$$fq-et=C_6+e(r-t)>0;\qquad fM-et>0\ \ (et<\tfrac{e^2M}{f}<fM).$$

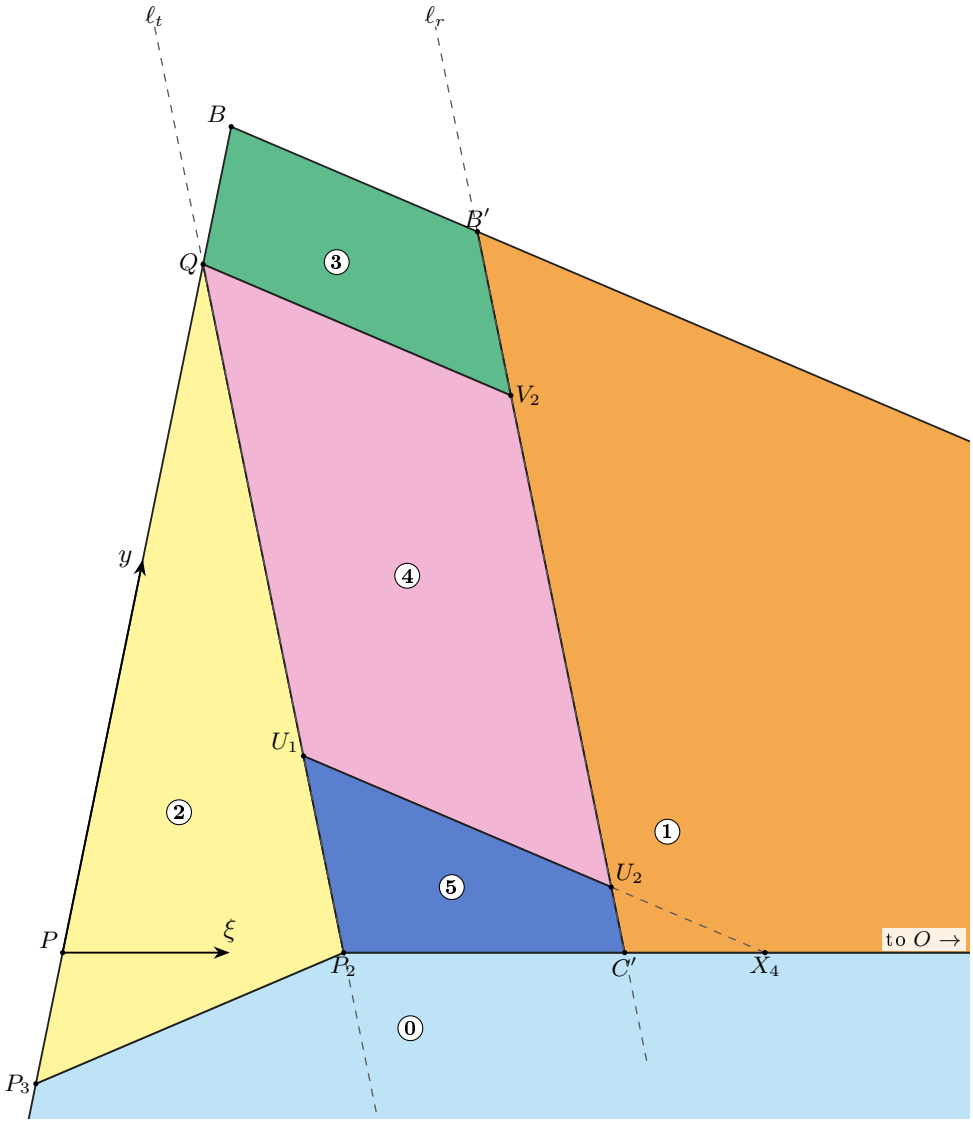


FIGURE 2. The same six pieces as in Figure 1 with the names of Tables 3 and 4 (true shape of the frame; O, A and R lie outside the window; P is the origin). Dashed: the lattice lines ℓ_t (through P_2, Q, U_1) and ℓ_r (through C', B', U_2, V_2), and the line through U_2, X_4. The point B lies between ℓ_t and ℓ_r on the side AB.

(The identity $eC_5 = \dots$ is checked by expanding.) Hence the six scales (2) are positive: $v = C_5$, $s_1 = f(r-t)$, $s_2 = C_4$, $x = C_7$, $u_1 = fq - et$, $u_2 = C_6$; moreover $s_1 - s_2 = C_3 > 0$ and $u_1 - u_2 = e(r-t) > 0$.

4.2. **Step 1: a chain lemma, and the pieces fill the target.** For points p, p', p'' in the plane the *turn* at p' is $\tau = (p'-p)\times(p''-p')$, where $(\xi, y)\times(\xi', y') = \xi y' - y\xi'$. The map from the frame to the true plane is linear with positive determinant, so the sign of every turn, orientation, convexity and winding numbers are the same in the frame and in the plane; below "counter-clockwise" refers to the frame. For a polygon with vertex list $p_1, \dots, p_m$, put $\partial\Pi = [p_1, p_2] + \dots + [p_m, p_1]$, a formal sum of oriented segments $[p, q]$ (from p to q), which we compute with two rules: (i)

$[p,q]+[q,p]=0$; (ii) $[p,q]+[q,r]=[p,r]$ for collinear p,q,r in any order. Two such sums are equal if they agree after applying these rules on each line.

Lemma 4.1 (chain lemma; [PROVED BY HAND]). *Let T and $\Pi_1,\dots,\Pi_k$ be triangles or quadrilaterals with counter-clockwise vertex lists all of whose turns are strictly positive. Each Π_i is then convex. If $\sum_i \partial\Pi_i = \partial T$ in the sense above, then the Π_i lie in T, have pairwise disjoint interiors, and cover T.*

Proof. A polygon with $m\le 4$ vertices and all turns positive has exterior angles in $(0,\pi)$ summing to a multiple of 2π that is less than $m\pi\le 4\pi$, hence to 2π; so its boundary is a simple closed convex curve. (For five or more vertices positive turns do not imply convexity; this is why only $m\le 4$ occurs.) The winding number of $\partial\Pi$ about a point z off the boundary is 1 if z is inside the convex counter-clockwise polygon and 0 outside. Winding numbers add over formal sums, so $\sum_i \mathbf{1}_{\Pi_i}(z)=\mathbf{1}_T(z)$ for all z off finitely many lines. If two interiors met, the left side would be ≥ 2 on an open set; if some Π_i left T it would be ≥ 1 outside T; and if some point of T were uncovered the right side would exceed the left. Interior and closure statements follow from convexity. □ □

(Caution: "every piece lies in T and the areas add up" does *not* imply disjointness; and $\sum\partial\Pi_i$ is a sum of segments, not of vectors.)

The turns. Take $T=OBA$ and the six pieces of Table 4. With $b=f^2-e^2$ the cross products are the following (each is a polynomial in e,f,M,t,r,q, factored; this is a finite expansion that the reader can repeat from Table 3, for instance Π_5 at P_2 is worked in Section 5):

polygon	turns at the vertices (in order)	value
$T=O,B,A$	all three	$M^2\,efb\,\kappa$
$\Pi_0=A,O,P_2,P_3$	A,O; P_2,P_3	$Mef^2b\,(fM-et)$; $e^2f\,bt\,(fM-et)$
$\Pi_1=O,B',C'$	all three	$efb\,(fM-er)^2$
$\Pi_2=P_3,P_2,Q$	all three	ef^3bt^2
$\Pi_3=B,Q,V_2,B'$	Q,V_2; B',B	$ef^2b\,(r-t)(eM-ft)$; $efb\,(fr-eM)(eM-ft)$
$\Pi_4=Q,U_1,U_2,V_2$	all four	$ef^2b\,(r-t)(ft-eq)$
$\Pi_5=U_1,P_2,C',U_2$	P_2,U_1; C',U_2	$e^2f\,b\,(r-t)(fq-et)$; $e^2f\,b\,(r-t)(fq-er)$

All of these are positive monomials times products of the nine numbers of Section 4.1, hence strictly positive under (K): the nine numbers are t, $r-t$, $eM-ft$, $fr-eM$, $fM-er$, $fM-et$, $ft-eq$, $fq-et$, $fq-er$. In particular T and the six pieces satisfy the hypothesis on turns.

The chain identity. Five lines carry all edges. With $\varphi=f\xi+ey$ and $\psi=e\xi+fy$: $AB=\{\xi=0\}$ contains A,P_3,Q,B,R; $OP=\{y=0\}$ contains O,P_2,C',X_4; $OB=\{\psi=efbM\}$ contains O,B,B'; $\ell_t=\{\varphi=efbt\}$ contains P_2,Q,U_1; $\ell_r=\{\varphi=efbr\}$ contains C',B',U_2,V_2. (For instance $\varphi(U_1)=f\cdot ef(ft-eq)+e\cdot ef(fq-et)=ef(f^2-e^2)t$, and $\psi(B')=e\cdot ef(fr-eM)+f\cdot ef(fM-er)=efbM$; these hold identically in e,f,M,t,r,q.) The edges of the pieces, line by line, are:

line	oriented edges of the pieces on it (sum)
AB	$[B,Q]_{\Pi_3}+[Q,P_3]_{\Pi_2}+[P_3,A]_{\Pi_0}=[B,A]$ (= the edge BA of T)
OB	$[O,B']_{\Pi_1}+[B',B]_{\Pi_3}=[O,B]$ (= the edge OB of T)
OA	$[A,O]_{\Pi_0}$ (= the edge AO of T)
OP	$[O,P_2]_{\Pi_0}+[P_2,C']_{\Pi_5}+[C',O]_{\Pi_1}=0$
ℓ_t	$[P_2,Q]_{\Pi_2}+[Q,U_1]_{\Pi_4}+[U_1,P_2]_{\Pi_5}=0$
ℓ_r	$[B',C']_{\Pi_1}+[C',U_2]_{\Pi_5}+[U_2,V_2]_{\Pi_4}+[V_2,B']_{\Pi_3}=0$
cuts	$[P_2,P_3]_{\Pi_0}+[P_3,P_2]_{\Pi_2}=0$, $[Q,V_2]_{\Pi_3}+[V_2,Q]_{\Pi_4}=0$, $[U_1,U_2]_{\Pi_4}+[U_2,U_1]_{\Pi_5}=0$

These are all 22 edges of the six pieces. On each line the sum is computed by rule (ii) (for example on AB: $[B,Q]+[Q,P_3]=[B,P_3]$ and $[B,P_3]+[P_3,A]=[B,A]$), and the three cut lines cancel by rule (i). So $\sum_i \partial\Pi_i = \partial T$.

Proposition 4.2 (Step 1; [PROVED BY HAND]). *If (K) holds, the six pieces $\Pi_0,\dots,\Pi_5$ lie in $W_M = OAB$, have disjoint interiors and cover it.*

Proof. Lemma 4.1 with the two displays above. □

4.3. **Step 2: each piece is a union of tiles.** By (1) the length of an integer multiple k of a primitive vector is k times the length in the table of Section 3.1. Table 5 lists the three sides of each of eight triangles as $k\cdot$(primitive vector) (each line is a subtraction of two points of Table 3); by SSS the triangle has the stated scale. For Π_4 two further identities: $U_1 - Q = fx\,(e,-f) =$

TABLE 5. The 24 edge identities: side vectors, and lengths in the form (scale)$\times(a$, b or $c)$.

triangle (scale)	side vectors	lengths
OAP (fM)	$O-P = fbM(1,0)$; $P-A = ef^2M(0,1)$; $A-O = -fM(b,ef)$	$b\cdot fM$; $a\cdot fM$; $c\cdot fM$
PP_2P_3 (et)	$P_2-P = bet(1,0)$; $P-P_3 = e^2ft(0,1)$; $P_3-P_2 = -et(b,ef)$	$b\cdot et$; $a\cdot et$; $c\cdot et$
P_3P_2Q (ft)	$Q-P_3 = f^3t(0,1)$; $Q-P_2 = -bt(e,-f)$; $P_2-P_3 = et(b,ef)$	$c\cdot ft$; $b\cdot ft$; $a\cdot ft$
$OB'C'$ (v)	$C'-O = -bv(1,0)$; $B'-O = -fv(f,-e)$; $B'-C' = -ev(e,-f)$	$b\cdot v$; $c\cdot v$; $a\cdot v$
RQV_2 (s_1)	$R-Q = fb(r-t)(0,1)$; $V_2-Q = es_1(f,-e)$; $V_2-R = fs_1(e,-f)$	$b\cdot s_1$; $a\cdot s_1$; $c\cdot s_1$
RBB' (s_2)	$R-B = bs_2(0,1)$; $B'-R = fs_2(e,-f)$; $B'-B = es_2(f,-e)$	$b\cdot s_2$; $c\cdot s_2$; $a\cdot s_2$
$X_4U_1P_2$ (u_1)	$P_2-X_4 = -bu_1(1,0)$; $U_1-X_4 = -fu_1(f,-e)$; $P_2-U_1 = eu_1(e,-f)$	$b\cdot u_1$; $c\cdot u_1$; $a\cdot u_1$
$X_4C'U_2$ (u_2)	$C'-X_4 = -bu_2(1,0)$; $U_2-X_4 = -fu_2(f,-e)$; $U_2-C' = -eu_2(e,-f)$	$b\cdot u_2$; $c\cdot u_2$; $a\cdot u_2$

$U_2 - V_2$ and $V_2 - Q = es_1(f,-e) = U_2 - U_1$ $(x = ft - eq)$, so Π_4 is a parallelogram with sides of lengths $c\,x$ and $a\,s_1$.

We use the ladder subdivision of the earlier release (Lean there, `Subdivision.ladderDissection`, `Ladder.ladder`).

Proposition 4.3 (ladder; [PROVED IN LEAN]). *A triangle of scale s is tiled by s^2 tiles. If a line parallel to a side cuts off from a triangle of scale s_1 the corner at the opposite vertex, which has scale $s_2 < s_1$, the remaining trapezoid is tiled by $s_1^2 - s_2^2$ tiles. A parallelogram with sides sa and uc enclosing the angle β is tiled by $2su$ tiles.*

Why. The lines parallel to the sides at integer steps cut a triangle of scale s into s^2 scale-one triangles, each congruent to the tile; a corner of scale s_2 at a vertex is s_2^2 of these cells. A parallelogram with sides a, c and angle β is cut by its short diagonal, of length b since $\cos\beta = \frac{a^2+c^2-b^2}{2ac}$, into two tiles; one with sides sa, uc is su of those.

Consequences (Table 4). (a) Π_0: OAP has scale fM; $P_2 \in PO$, $P_3 \in PA$ (as $et \le fM$), and $P_2P_3 \parallel OA$ (direction (b,ef)), so PP_2P_3 is its corner at P, of scale et: $f^2M^2 - e^2t^2$ tiles. (b) Π_1, Π_2 are triangles of scales v and ft. (c) Π_3: R lies on AB beyond B $(ft < eM < fr)$,

$B \in RQ$, $B' \in RV_2$, $BB' \parallel QV_2$ (both direction $(f,-e)$), so $\Pi_3 = RQV_2 \setminus RBB'$: $s_1^2 - s_2^2$ tiles. (d) Π_5: $X_4U_1P_2$ has scale u_1 and $C'U_2 \parallel P_2U_1$ cuts off the corner $X_4C'U_2$ of scale u_2: $u_1^2 - u_2^2$ tiles. (e) Π_4: the angle at Q between $(f,-e)$ and $(e,-f)$ has cosine, by (1),

$$\frac{\langle (f,-e),(e,-f)\rangle}{f\cdot f} = \frac{2ef - \frac{e}{2f}(f^2+e^2)}{f^2} = \frac{e(3f^2-e^2)}{2f^3} = \cos\beta,$$

so Π_4 is a parallelogram with sides as_1, cx and angle β: $2s_1x$ tiles.

4.4. **The number of tiles.** By Proposition 4.2 and the consequences above, W_M is tiled; it remains to see that the number of tiles is $M^2\kappa$. The turn of a triangle of scale s is $s^2\cdot efb$ (compare the table of turns: Π_1, Π_2 give $efb\,v^2$, $efb(ft)^2$, and the triangle O,P,A of scale fM has turn $(P-O)\times(A-P) = efb(fM)^2$), because the area is s^2 times the tile's, and the turn of a triangle (any vertex) equals $2\cdot\text{area}/\sin\theta_P$ where $\theta_P = \arccos\frac{e}{2f}$ is the angle of the frame, the same constant for every triangle. The turn of T is $M^2efb\,\kappa$, hence T has area $M^2\kappa$ tiles. Since all tiles have the same area, the number of tiles is $M^2\kappa = M^2(2f^2-e^2)$. (Consistency: the six counts of Table 4 add up to $M^2\kappa$, an identity in e,f,M,t,r,q; for $W_3(2,5)$: $221+121+25+9+30+8 = 414$.)

4.5. **Step 3: the canonical parameters.** Put $n = \lfloor eM/f\rfloor$, $\rho = eM - fn \in \{0,\dots,f-1\}$, $\delta = 1$ if $\rho = 0$ and $\delta = 0$ otherwise.

Lemma 4.4 (Lemma H; [PROVED BY HAND])**.** *Let* $\gcd(e,f) = 1$, $2 \le e \le f-2$, $d = f-e$, $M \ge 1$. *Then (H1) and (H2) hold if and only if* $dn \ge e+1$.

Proof. First, $ed \ge f+1$: with $f = e+d$, $ed - f = (e-1)(d-1) - 1$, and $(e-1)(d-1) \ge 2$ because $e,d \ge 2$ and $e = d = 2$ would make $f = 4$ not coprime to e. $(*)$

($\Leftarrow$) Let $dn \ge e+1$. Then $n \ge 1$, i.e. (H1). If (H2) failed, $(M-1)d \le f-1$, so $dM \le f+d-1 < 2f$ and $s := \lceil dM/f\rceil \in \{1,2\}$; $n = \lfloor M - dM/f\rfloor = M - s$. If $s = 1$: $dM \le f$, so $dn = d(M-1) \le f-d = e$. If $s = 2$: $dn = d(M-2) \le f+d-1-2d = e-1$. Either way $dn \le e$, a contradiction.

($\Rightarrow$) Let (H1), (H2). If $d \ge e+1$: $n \ge 1$ by (H1), so $dn \ge d \ge e+1$. If $d \le e$: $d \nmid f$ (else $d \mid e$, so $d = 1$, excluded), so $f = dh + j$ with $0 < j < d$, and $h \ge 2$ as $f = e+d \ge 2d$; $e = d(h-1)+j$. (H2) gives $M - 1 \ge \lceil f/d\rceil = h+1$, so $M \ge h+2$ and $e(h+2) - hf = 2e - hd = d(h-2) + 2j > 0$. So $eM \ge e(h+2) > hf$, $n \ge h$, and $dn \ge dh = f - j \ge f - d + 1 = e+1$. □

Proposition 4.5 (the five strict inequalities, with certificates; [PROVED BY HAND])**.** *Assume (H1), (H2) and put* $t = n-\delta$, $r = n+1$, $q = n$, $s = M-n$ (≥ 1), $\mu = dn - e - 1$, $\rho' = f - 1 - \rho$. *Then* $\mu \ge 0$, $\rho' \ge 0$ *and*

$$C_3 = \rho + f\delta, \qquad C_4 = \rho' + 1, \qquad C_5 = \mu + 1 + fs, \qquad C_6 = \mu + 1,$$

$$C_7 = \begin{cases} \mu + 1 + e & \delta = 0, \\ (m'-1)ed + \big((e-1)(d-1) - 1\big) & \delta = 1, \end{cases}$$

where for $\delta = 1$ *we write* $M = fm'$, $n = em'$. *Each is positive, so (K) holds.*

Proof. $\mu \ge 0$ is Lemma 4.4. $C_3 = eM - f(n-\delta) = \rho + f\delta$, positive since $\rho \ge 1$ if $\delta = 0$; $C_4 = f(n+1) - eM = f - \rho = \rho' + 1$; $C_6 = fn - e(n+1) = dn - e = \mu + 1$; $C_5 = f(n+s) - e(n+1) = dn - e + fs = \mu + 1 + fs$. $C_7 = f(n-\delta) - en = dn - f\delta$: if $\delta = 0$ it is $\mu + 1 + e$; if $\delta = 1$ then $f \mid eM$ gives $f \mid M$ (coprimality), $M = fm'$, $n = em'$, and $dn - f = (m'-1)ed + (ed - f)$ with $ed - f = (e-1)(d-1) - 1 \ge 1$ by $(*)$. Every term is non-negative and each total contains a positive constant. □

End of the proof of the W half. For the canonical (t, r, q) condition (K) holds (Proposition 4.5); Proposition 4.2 places the six pieces; Section 4.3 tiles them; the count is $M^2(2f^2 - e^2)$. The T half is Proposition 2.3. □

Remark 4.6 (where the hypotheses enter; the parameters)**.** (H1) gives $n \geq 1$; (H2) enters only through $dn \geq e + 1$, which is $C_6 > 0$ for $q = n$, $r = n + 1$: it says that the apex X_4 of the triangle from which the corner of Π_5 is cut lies on PO strictly beyond C' (indeed $PX_4 - PC' = fbq - ber = b\,C_6$).

The triple (t, r, q) is not unique: for $W_3(2, 5)$ the triples $(1, 2, 1), (1, 2, 2), (1, 3, 2), (1, 4, 2)$ all satisfy (K). What $ft < eM < fr$ forces is $t \leq n - \delta$ and $r \geq n + 1$. Moreover, if (K) has any solution then $dn \geq e+1$. Indeed, suppose $dn \leq e$, i.e. $fn \leq e(n+1)$. Then $fq > er \geq e(n+1) \geq fn$, so $q \geq n + 1$, and then $eq \geq e(n + 1) \geq fn \geq ft > eq$, a contradiction. Hence the layout exists for some integers t, r, q exactly when $dn \geq e + 1$ (Lemma 4.4 and Proposition 4.5). This is a statement about this layout, not about tilings.

5. Worked rows for $W_3(2, 5)$

Example 5.1 ($(e, f, M) = (2, 5, 3)$, 414 tiles)**.** $a = 10$, $b = 21$, $c = 25$, $\kappa = 46$, $d = 3$. W_3 has sides $Mfb = 315$, $Me\kappa = 276$, $Mf^3 = 375$. $n = \lfloor 6/5 \rfloor = 1$, $\rho = 1$, $\delta = 0$, $(t, r, q) = (1, 2, 1)$. The five numbers: $C_3 = eM - ft = 1$, $C_4 = 4$, $C_5 = 11$, $C_6 = 1$, $C_7 = 3$ (certificates: $\mu = dn - e - 1 = 0$, $\rho' = 3$, $s = 2$: $C_3 = \rho = 1$, $C_4 = \rho' + 1 = 4$, $C_5 = \mu + 1 + fs = 11$, $C_6 = 1$, $C_7 = \mu + 1 + e = 3$). Scales: $v = 11$, $s_1 = 5$, $s_2 = 4$, $x = 3$, $u_1 = 3$, $u_2 = 1$. Points (ξ, y): $P = (0, 0)$, $O = (315, 0)$, $B = (0, 126)$, $A = (0, -150)$, $P_2 = (42, 0)$, $P_3 = (0, -20)$, $Q = (0, 105)$, $R = (0, 210)$, $C' = (84, 0)$, $B' = (40, 110)$, $V_2 = (50, 85)$, $X_4 = (105, 0)$, $U_1 = (30, 30)$, $U_2 = (80, 10)$. *Line equations* (Step 1): $\varphi = 5\xi + 2y$: P_2: 210, Q: 210, U_1: 210 $(= efbt)$; C': 420, B': 420, U_2: 420, V_2: 420 $(= efbr)$. $\psi = 2\xi + 5y$: O: 630, B: 630, B': 630 $(= efbM)$. *Turns* (Step 1), $\Pi_5 = U_1, P_2, C', U_2$: edge vectors $U_1 \to P_2 = (12, -30)$, $P_2 \to C' = (42, 0)$, $C' \to U_2 = (-4, 10)$, $U_2 \to U_1 = (-50, 20)$; turns $12 \cdot 0 - (-30) \cdot 42 = 1260$, $42 \cdot 10 - 0 = 420$, $(-4) \cdot 20 - 10 \cdot (-50) = 420$, $(-50)(-30) - 20 \cdot 12 = 1260$. The table gives $e^2fb(r-t)(fq-et) = 4 \cdot 5 \cdot 21 \cdot 1 \cdot 3 = 1260$ at P_2, U_1 and $e^2fb(r-t)(fq-er) = 4 \cdot 5 \cdot 21 \cdot 1 \cdot 1 = 420$ at C', U_2. Another row: Π_4 at Q: $efbf(r - t)(ft - eq) = 10 \cdot 21 \cdot 5 \cdot 1 \cdot 3 = 3150$, and directly $(U_1 - Q) \times (U_2 - U_1) = (30, -75) \times (50, -20) = 30(-20) - (-75)(50) = 3150$. *An edge identity* (Step 2): $P_2 - U_1 = (12, -30) = eu_1(e, -f) = 6\,(2, -5)$, of length $6f = 30 = a\,u_1$. *Counts:*

	scales	formula	tiles		scales	formula	tiles
Π_0	15 minus 2	$225 - 4$	221	Π_3	5 minus 4	$25 - 16$	9
Π_1	11	11^2	121	Π_4	$as_1 \times cx$	$2 \cdot 5 \cdot 3$	30
Π_2	5	5^2	25	Π_5	3 minus 1	$9 - 1$	8

Total $414 = 9 \cdot 46$. These numbers are the edge labels and counts of Figure 1.

Example 5.2 ($W_4(3, 5)$, 656 tiles: scale $4 < 5$)**.** $(e, f, M) = (3, 5, 4)$, $a = 15$, $b = 16$, $c = 25$, $\kappa = 41$, $n = 2$, $\rho = 2$, $\delta = 0$, $(t, r, q) = (2, 3, 2)$; scales $v = 11$, $s_1 = 5$, $s_2 = 3$, $x = 4$, $u_1 = 4$, $u_2 = 1$; counts $364 + 121 + 100 + 16 + 40 + 15 = 656 = 16 \cdot 41$.

Example 5.3 (the case $\delta = 1$: $(2, 5, 5)$, $M = f$)**.** $n = 2$, $\rho = 0$, $(t, r, q) = (1, 3, 2)$, $C_3 = 5$, $C_4 = 5$, $C_6 = 4$, $C_7 = 1$; scales $v = 19$, $s_1 = 10$, $s_2 = 5$, $x = 1$, $u_1 = 8$, $u_2 = 4$; counts $621 + 361 + 25 + 75 + 20 + 48 = 1150 = 25 \cdot 46$.

Example 5.4 (no layout: $(3, 5, 3)$)**.** $d = 2$, $n = 1$, $dn = 2 \leq 3 = e$, so (K) has no solution (Remark 4.6): the canonical triple $(1, 2, 1)$ has $C_6 = fq - er = 5 - 6 = -1 < 0$. This does not show that $W_3(3, 5)$ is untiled.

6. What is known and what is open

Known. W_M is tiled if $M \geq M^*$ (Theorem 2.1) and not tiled if $Me < f$ (I1) or $M(f-e) \leq e$ (I2), both imported; for $e < f/2$ this is a sharp threshold (Corollary 2.4); for $e = 1$ and $e = f-1$ the threshold is f (I3). For $e > f/2$ the scales neither covered by Theorem 2.1 nor excluded by (I2) are those with $e/d < M < f/d+1$ ($d = f-e$), an open interval of length 2, so at most two per row. For T_M the scales with $2Me \geq f$ and $M < M^*$ are neither proved tiled nor excluded (for example $T_3(2,11)$).

Computed, not proved. $W_2(5,8)$ (412 tiles) is the smallest such open case with $e > f/2$ we examined: an exact search which imposes one of 32 base-edge words and assumes a search parameter ($r = 6$) found no tiling for any of the 32 words. This is evidence only: it is not a proof (the assumption and the restriction to those words are not justified), and it is not formalised.

Conjecture 6.1 (W^*; [conjecture])**.** *For coprime $2 \leq e \leq f-2$, $W_M(e,f)$ is tiled if and only if $M \geq M^*(e,f)$.*

Corollary 2.4 gives it for $e < f/2$ (with (I1)). Nothing here concerns the prime case or any integer other than those of Section 7.1.

7. Effect on the census, credits, disclosure

7.1. **The census.** Let F be the set of the numbers $M^2(2f^2-e^2)$ and $M^2(3f^2-e^2)$, over the triples (e,f,M) of Theorem 2.1, that are non-squarefree and at most 10^6. By the theorem they are all tile counts. The census of the companion paper [6] lists, for the non-squarefree integers $2 \leq N \leq 10^6$, which are known tile counts, which are excluded and which are undecided. Relative to that census F has 44 495 elements (28 946 of the form W, 18 431 of the form T, 2 882 of both; largest f is 251); none is excluded by the exclusion theorem of the earlier release; 18 357 were known tile counts by other constructions. So 26 138 **integers become tile counts of the form $M^2(2f^2-e^2)$ or $M^2(3f^2-e^2)$ relative to that census**: they moved from "undecided" to "tile count". The count 26 138 comes from a program (computed, exact), and "undecided" is the status in the companion's census, not a statement about the literature. We found no statement of these values by Paliy or Beeson–Roşu (a lookup in Paliy's repository at commit `0f7935b` only and in the cited Beeson texts; not a proof that none exists). Examples: $414 = W_3(2,5)$ and $639 = T_3(2,5)$ were undecided before.

7.2. **Credit.** The targets W_M, T_M, the W/β framework and the tilings for $M \geq f$ are due to D. Paliy and to M. Beeson and G. Roşu [2] (the tile with $3\alpha + 2\beta = \pi$ is Beeson's case III [1]); Paliy's up-set in the scale was published on 29 September 2026; the isosceles-apex frame at the point P with $OP = OB$ is that of Beeson–Roşu. The six-piece layout was found by our own computer search; the integer coordinates, the chain (winding-number) lemma and Lemma H come from sessions inside this project. The companion paper [6] credits B. Herdt, Y. X. Zhang, M. Laczkovich, J. P. Harries and Snover–Waiveris–Williams, whose work enters there. Statements about Paliy's work refer to his public repository [3]; nothing here claims priority over him for what is listed as his.

7.3. **Disclosure.** The work was done with the help of an AI system (Claude, Anthropic); the questions and directions were the author's; each statement carries a label saying how it was checked.

References

[1] M. Beeson, *Triangle tiling III: the case* $3\alpha + 2\beta = \pi$, arXiv:1206.2229.

[2] M. Beeson and G. Roşu, *Tiling a triangle into a prime number of congruent triangles*, preprint, version of 16 October 2026.
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[4] V. Bonfioli, *Congruent dissections of a triangle: prime counts, the counts up to* 200*, and a family of isosceles targets*, release v1.13.1, Zenodo, October 2026 (concept DOI doi:10.5281/zenodo.23400433).
[5] V. Bonfioli, *Congruent dissections of a triangle: runs, windows and the corner*, version 2.2, release v1.16, Zenodo (same concept DOI).
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[7] V. Bonfioli, *Paper W: verification record* (a separate short document in the same release): what was computed how, the older complex-coordinate proof, data and Lean index.